

# Aviator Quant Bot — Chrome

## Extension

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Adaptive, probability-based strategy bot for the Aviator game (by Spribe). Uses quantitative trading principles — Kelly Criterion sizing, expected value filtering, volatility guards, and streak detection — to make mathematically honest betting decisions.

**This is NOT prediction. It is adaptive risk management based on live session data.**

## Install

1. Open Chrome → `chrome://extensions/`
2. Enable **Developer mode** (top-right toggle)
3. Click **Load unpacked** → select the `aviator-strategy-bot` folder
4. Navigate to any site running the Aviator game
5. Click the extension icon to show the panel

## Core Philosophy

The bot cannot predict crash points. But it CAN:

1. Track a rolling window of recent crash points
2. Calculate real-time statistics from that data
3. Dynamically adjust bet size and cashout target
4. Only bet when expected value is favorable
5. Cut exposure during high-variance periods

# How the Algorithm Works

## Module 1 — Crash History Tracker

Maintains a **rolling window of the last 50 crash points**. After every round, it calculates:

- **Mean** — average crash value
- **Standard Deviation** — measure of volatility
- **% above 1.5x** — how often rounds exceed 1.5x
- **% above 2x** — how often rounds exceed 2x
- **% above 3x** — how often rounds exceed 3x
- **Recent low streak** — consecutive crashes below 1.5x (danger signal)

The bot waits for **at least 10 data points** before making any bet decisions. During this collection phase it only observes.

## Module 2 — Kelly Criterion Bet Sizing

Used by professional gamblers and quant funds to size positions based on edge and probability.

$$\text{Kelly \%} = (p \times b - q) / b$$

Where:

p = probability of winning (from history)

q = 1 - p (probability of losing)

b = net odds (cashout multiplier - 1)

The bot uses **Half-Kelly** (standard quant practice — reduces variance at the cost of slightly lower returns). Bet size is capped at **5% of bankroll maximum**. If Kelly returns zero or negative, the round is skipped entirely — negative Kelly means the bet has no edge.

## Module 3 — Dynamic Cashout Target

The bot does NOT use a fixed target. It picks the optimal cashout each round based on what the data supports:

| Condition                           | Target | Logic                          |
|-------------------------------------|--------|--------------------------------|
| 55%+ of recent rounds went above 2x | 2.0x   | Session is running hot         |
| 55%+ went above 1.5x                | 1.5x   | Normal conditions              |
| 45%+ went above 1.5x                | 1.3x   | Session is cold — conservative |
| Below 45%                           | Skip   | No edge exists                 |

## Module 4 — Volatility Guard

When crash points are highly variable (high standard deviation), the bot reduces exposure:

| StdDev  | Volatility Level | Action                 |
|---------|------------------|------------------------|
| < 1.5   | Low              | Normal bet size (100%) |
| 1.5–3.0 | Medium           | Reduce bet by 40%      |
| 3.0–5.0 | High             | Reduce bet by 70%      |
| > 5.0   | Extreme          | Skip round entirely    |

## Module 5 — Streak Detection

The bot adjusts behavior based on consecutive outcomes:

| Condition                    | Response                             |
|------------------------------|--------------------------------------|
| 4+ consecutive losses        | Reduce bet to 50% (never martingale) |
| 6+ consecutive losses        | Skip round entirely                  |
| 3+ recent crashes below 1.5x | Skip round                           |
| 4+ consecutive wins          | Hold steady — do NOT increase bets   |

**Critical rule:** The bot NEVER uses Martingale (doubling after loss). Ever.

## Module 6 — Master Decision Engine

Before every round, combines all modules into a single decision:

1. Calculate rolling stats from history
2. Determine dynamic cashout target
3. Check if target has positive EV:  $EV = (winProb \times payout) - lossProb$
4. If  $EV \leq 0 \rightarrow$  skip
5. Calculate Kelly bet size for that target
6. Apply volatility multiplier
7. Apply streak adjustment
8. Final sanity checks (minimum bet, bankroll cap)

If the bot skips 3 **consecutive rounds**, it forces a minimum 1% bet at 1.5x to keep collecting data.

## Module 7 — Stop Conditions

| Rule               | Threshold                       | Action    |
|--------------------|---------------------------------|-----------|
| Drawdown from peak | 25% drop from session high      | Bot stops |
| Take-Profit        | 40% gain from starting bankroll | Bot stops |
| Max Rounds         | Configurable (default 100)      | Bot stops |
| Bankrupt           | Bankroll reaches ₹0             | Bot stops |

Note: Stop-loss is measured from **session peak** (highest bankroll reached), not just the starting amount. This is a trailing stop.

## Available Strategies (9 total)

The bot implements 9 distinct betting strategies. Each has a fundamentally different approach to bet sizing, target selection, and risk management.

### 1. Quant/Kelly (default)

Uses Kelly Criterion to size bets based on detected edge. Dynamically picks cashout target (1.1x–2.0x) from rolling 50-round history. Skips rounds when no positive expected value is found. Dual-panel split: 70% safe + 30% hunter.

## 2. Goal Seeker (Dubins-Savage Bold Play)

**Framework:** Dubins-Savage Bold Play + Browne's Goal-Reaching.

Dynamically adjusts aggression based on where you stand between your stop-loss floor and profit-target ceiling. The closer you are to ruin, the bolder the bets — the closer you are to your goal, the smaller and safer.

### Aggression Zones:

| Zone     | Bankroll Position        | Behavior                                   |
|----------|--------------------------|--------------------------------------------|
| TARGET   | > 95% toward profit goal | Lock-in mode: tiny bets, 1.2x cashout      |
| COMFORT  | 70–95% toward goal       | Conservative: small bets, 1.5x target      |
| NEUTRAL  | 30–70% toward goal       | Standard: Kelly-based sizing               |
| DANGER   | 10–30% toward goal       | Aggressive: bigger bets, higher targets    |
| CRITICAL | < 10% from stop-loss     | Desperation: bold play, max payout targets |

**Key math:** - Gambler's Ruin probability computed in real-time:  $P(\text{reach target}) = \frac{(1 - (q/p)^{\text{current}})}{(1 - (q/p)^{\text{target}})}$  - Bet sizes and targets derived from distance-to-goal, NOT from crash history - Completely stateless — all decisions derived from current bankroll position relative to boundaries

**Best for:** Sessions with a clear profit target and stop-loss, where you want the bot to "fight" its way to the goal with calibrated aggression.

## 3. Sniper (Adaptive)

A multi-phase "observe then fire" strategy. Fully autonomous — no manual target or window configuration needed.

### Lifecycle:

```
INITIAL_OBSERVE (50 rounds) → Analysis → READY → [trigger check] → FIRE
    ↓ (after each bet)
OBSERVE (50 rounds) → Re-analysis → READY → [trigger check] → FIRE
    ↓ (repeat until window exhausted or hit)
DONE → restart from INITIAL_OBSERVE
```

**Phase 1 — Observation (50 rounds, configurable):** The bot watches crash data without betting. It analyzes 6 multiplier tiers (10x, 15x, 20x, 30x, 50x, 100x) and scores each by: - Observed frequency vs. theoretical expectation - Edge (observed rate – theoretical rate) - Payout ratio

**Phase 2 — Window Sizing:** After analysis, the bot picks the "hottest" tier and calculates how many bets (window size) are needed to achieve a target cumulative hit probability:

```
window_size = ceil(ln(1 - confidence) / ln(1 - P(hit per round)))
```

Example at 15x target ( $P \approx 6.5\%$  per round):

| Confidence         | Window Size | Cost (at ₹10/bet) | P(at least 1 hit) |
|--------------------|-------------|-------------------|-------------------|
| 70% (Conservative) | ~18 bets    | ₹180              | 70%               |
| 50% (Balanced)     | ~10 bets    | ₹100              | 50%               |
| 30% (Aggressive)   | ~5 bets     | ₹50               | 30%               |

**Phase 3 — Trigger (Wait Condition):** Before firing, the bot can optionally wait for a signal:

| Trigger   | Condition                         | Rationale                          |
|-----------|-----------------------------------|------------------------------------|
| Immediate | Fire on next round                | Default — no delay after analysis  |
| Dry Spell | Last 5 rounds ALL below 2x        | Patience: wait for a "cold" streak |
| Overdue   | Target not seen in last 20 rounds | Patience: target feels "due"       |

Note: Dry Spell and Overdue do NOT improve mathematical odds (each round is independent). They add discipline/patience by delaying entry.

**Phase 4 — Fire:** Places the bet with calculated amount and target. If dynamic target is enabled (default), re-observes 50 rounds before EACH individual bet in the window and re-analyzes — the target may shift between bets.

**Configurable Parameters:**

| Setting             | Default   | Description                                         |
|---------------------|-----------|-----------------------------------------------------|
| Observation Rounds  | 50        | How many rounds to watch before each bet            |
| Max Risk per Window | 5%        | Total bankroll % risked across all bets in a window |
| Window Confidence   | 50%       | Target P(at least 1 hit) — controls window size     |
| Dynamic Target      | On        | Recalculate target between bets within a window     |
| Fire Trigger        | Immediate | When to pull trigger after observation completes    |

**Best for:** High-multiplier hunting with patience. You accept many rounds of pure observation for occasional big payouts.

#### 4. Flat Conservative

Always bets 1% of bankroll. Always cashes out at 1.5x (64.7% win rate). Never changes. Maximum bankroll survival for long sessions with minimal drawdown.

#### 5. Flat Balanced

Fixed 1% of bankroll per round. Cashout at 2.0x (48.5% win rate). Balanced risk/reward — wins less often but profits more per win.

#### 6. Anti-Martingale

Positive progression: doubles bet after each win ( $1x \rightarrow 2x \rightarrow 4x$ ). Resets to base after any loss or 3 consecutive wins. Capitalizes on hot streaks safely. Never increases after losses.

#### 7. 1-3-2-6

Structured 4-step sequence on consecutive wins: bet 1, 3, 2, 6 units. Reset to step 1 on any loss. Profits from 4-win streaks (pays 12 units on 12-unit investment). Self-limiting.

#### 8. D'Alembert

Gentle negative progression: +1 unit on loss, -1 on win, never below 1 unit. Much safer than Martingale — linear growth instead of exponential. Slow recovery, high bankroll survival.

## 9. Oscar's Grind

Ultra-conservative cycle system. Only increases bet by 1 unit after a win AND only if the cycle is negative. Resets once cycle reaches +1 unit profit. Patient capital preservation.

## Auto-Switch Strategy

Evaluates performance every 25 rounds using a composite scoring system and switches to the best-performing strategy automatically.

**Rules:** - Minimum 15 rounds on a strategy before switching is considered - Excludes Goal Seeker and Sniper from rotation (they have dedicated lifecycles) - Scores based on: win rate, drawdown, EV, volatility-adjusted returns

## Panel Dashboard

The floating sidebar shows live quant data:

**Decision Engine section:** - EV (expected value of current decision) - Win Probability (from rolling history) - Dynamic Cashout Target - Kelly % being used - Volatility level (Low/Medium/High/Extreme) - Bet size for this round - Decision reason (why it bet or skipped)

**Rolling Stats section:** - Mean crash, StdDev - % above 1.5x, 2x, 3x - Sample count - Sparkline of last 20 crash points (color-coded: green >2x, orange >1.5x, red <1.5x)

**Bankroll section:** - Starting, Current, Peak, P/L

**Session section:** - Rounds played, Win rate, Skipped rounds, Consecutive losses/wins

## User Configuration

| Setting           | Default | Description                             |
|-------------------|---------|-----------------------------------------|
| Starting Bankroll | ₹1000   | Your initial capital                    |
| Max Rounds        | 100     | Auto-stop after this many rounds        |
| Drawdown Stop     | 5%      | Stop if bankroll drops this % from peak |



| Setting         | Default | Description                              |
|-----------------|---------|------------------------------------------|
| Take-Profit     | 110%    | Stop if bankroll reaches this % of start |
| Simulation Mode | On      | Dry run with generated crash points      |

## Simulation Mode

When enabled (default), the bot:

- Generates crash points with a weighted distribution mimicking real game behavior
- Runs full quant logic (Kelly, EV, volatility) against simulated data
- Shows timing that mirrors real game phases (betting → flying → crash → pause)
- Does NOT interact with the game DOM

Disable simulation only when you want the bot to evaluate against real detected crash multipliers.

## Files

```
manifest.json    - Chrome extension manifest (V3)
background.js    - Service worker for storage/lifecycle
content.js       - Main bot: game observation, decision loop, simulation
strategies.js    - All 9 strategy implementations + auto-switch logic
panel.js         - Floating dashboard UI + configuration
panel.css        - Panel styles
icons/          - Extension icons
```

## Mathematical Honesty

- The house edge is ~3% (provably fair RNG)
- NO strategy can guarantee long-term profit against a negative-expectation game
- Observation does NOT predict future rounds — each crash is independent
- "Hot" tiers, dry spells, and overdue signals are pattern-matching heuristics, not causal predictors
- What math CAN do: optimize bet sizing, manage risk, maximize P(reaching a goal before ruin)
- The Gambler's Ruin theorem proves that with infinite play, the house always wins

- These strategies optimize **discipline and risk management**, not luck

## Disclaimer

This is for educational and entertainment purposes. The house edge (~3%) means no strategy guarantees long-term profit. This bot minimizes bad decisions — it does not eliminate risk.